The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The roots of a quadratic polynomial

Derivative as a Limit

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

The slope of the tangent line

The Fundamental Theorem of Calculus

$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

Integration undoes differentiation

The Gaussian Integral

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

Area under the bell curve

Geometric Series

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

$$S_N = \frac{a(1-r^N)}{1-r}$$

The infinite sum needs |r| less than 1

Euler's Identity

$$e^{i\pi} + 1 = 0$$

Five constants in one line

A Two by Two Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Zero means the rows are dependent

A Linear System as a Matrix Equation

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

Written as A x = b

Newton's Second Law

$$F = ma$$

$$p = mv$$

Force, mass and acceleration

Mass and Energy

$$E = mc^2$$

Rest energy of a mass m

The Schrodinger Equation

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$$

Time evolution of a quantum state

The Ideal Gas Law

$$PV = nRT$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Pressure, volume and temperature

Photosynthesis

$$6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$$

Sunlight drives the reaction

Self-Ionisation of Water

$$2\,H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

A very small equilibrium constant

Radiocarbon Decay

$$^{14}_{6}\mathrm{C} \longrightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e} + \bar{\nu}_e$$

Beta decay with a half-life of 5730 years

Alpha Decay of Uranium

$$^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

Mass number and atomic number are conserved

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

Updating a probability with evidence

The Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mean mu and standard deviation sigma

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

A function as a power series about a

The Binomial Theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

Expanding a power of a sum

The Law of Cosines

$$c^2 = a^2 + b^2 - 2ab\cos C$$

A generalisation of Pythagoras

Logarithm of a Product

$$\log_b(xy) = \log_b x + \log_b y$$

Logarithms turn products into sums

Cramer's Rule

$$x = \frac{\begin{vmatrix} 5 & 1 \\ 10 & 3 \end{vmatrix}}{\begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix}}$$

Each unknown is a ratio of determinants

The Absolute Value Function

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Defined by cases

A System with a Brace

$$\begin{cases} 2x + y = 5 \\ x + 3y = 10 \end{cases}$$

Solve for x and y together

The Chain Rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Differentiating a composite function

Pythagoras

$$a^2 + b^2 = c^2$$

For a right-angled triangle

Combustion of Methane

$$\mathrm{CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(l)}$$

Carbon dioxide and water are the products

Precipitation of Silver Chloride

$$\mathrm{Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)}$$

The net ionic equation